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Roll no. ....

Degree: B.Tech., Semester: Third-Course work
END-SEMESTER THEORY EXAMINATION, NOV-DEC, 2024

Course Code: CMMTC08
Course Title: Mathematical Statistics

Time: 03 Hours

Max. Marks: 50

Note: Attempt all the five questions. Missing data/ information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO
Q1	Attempt any 2 parts of the following:		
1(a)	The pdf of a continuous random variable X is $$f(x) = \begin{cases} ke^{-x} & 0 < x < \infty \\ 0 & otherwise. \end{cases}$$ Find the value of k, mean, and variance.	5	CO1
1(b)	What is the Poisson Process? With the help of an example, derive the Poisson distribution.	5	CO1
1(c)	Let X be the number of births in a family until the second daughter is born. If the probability of having a male child is $1/2$, find the probability that the sixth child in the family is the second daughter.	5	CO1
Q2	Attempt any 2 parts of the following:		
2(a)	If the first four moments of a distribution about the value 5 are equal to $-4,\ 22,\ -117$ and 560, determine the corresponding moments about the mean.	5	CO2
2(b)	Subway trains on a certain line run every half hour between midnight and six in the morning. What is the probability that a man entering the station at a random time during this period will have to wait at least 20 minutes?	5	CO2
2(c)	State Chebyshev's inequality and Markov's inequality.	5	CO2
Q3	Attempt any 2 parts of the following:		
3(a)	Calculate the moment generating function of Poisson distribution.	5	CO3
3(b)	State and prove the Central Limit Theorem.	5	CO3
3(c)	Fit a parabola of the second degree to the following data: <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Y</td><td>1</td><td>1.8</td><td>1.3</td><td>2.5</td><td>6.3</td></tr></table>	5	CO3

Q4	Attempt any 2 parts of the following:		
4(a)	In a random sample of 500 men from a particular district of U.P., 300 are found to be smokers. In one of 1,000 men from another district, 550 are smokers. Do the data indicate that the two districts are significantly different with respect to the prevalence of smoking among men? (Use 5% level of significance)	5	CO4
4(b)	A die is thrown 60 times with the following results: Face: 1, 2, 3, 4, 5, 6 Frequency: 8, 7, 12, 8, 14, 11 Test at 5% level of significance if the die is fair. (Use $\chi^2(5, 0.05) = 11.1$)	5	CO4
4(c)	In an experiment with immunization of cattle from tuberculosis, the following results were obtained: Inoculated: Affected 12, Unaffected 26 Not Inoculated: Affected 16, Unaffected 6 Examine the effect of the vaccine in controlling the susceptibility of tuberculosis. (Use $\chi^2(1, 0.05) = 3.841$)	5	CO4
Q5	Attempt any 2 parts of the following:		
5(a)	Write the probability density function of χ^2, F, t-distributions, and relationship between these distributions.	5	CO5
5(b)	A random sample of size 16 from a normal population showed a mean of 41.5 inches and the sum of squares of deviations from this mean equal to 135 square inches. Can this sample be regarded as taken from the population having 43.5 as mean? (Use $t_{15}(0.05) = 2.131$)	5	CO5
5(c)	Find mean and variance of F-distribution.	5	CO5

Table for 4(b):

Face	1	2	3	4	5	6
Frequency	8	7	12	8	14	11

Table for 4(c):

	Affected	Unaffected
Inoculated	12	26
Not Inoculated	16	6